

Bragg Diffraction with Sodium Chloride and Lithium Fluoride Crystals

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History and Background (CK, MN)

Before Lawrence and William Bragg won the 1915 Nobel Prize as a father-son team for their work with crystallography and X-rays, chemists and physicists believed that materials like salts were made up of individually bound molecules. The work of these English physicists revealed that such materials actually exhibit a crystal-like lattice structure composed of alternating positively and negatively charged ions¹. This was the birth of X-ray crystallography. Lawrence Bragg developed a crucial idea that explained that the manner in which X-rays get reflected by crystals is due to a layering of planes of atoms within the crystal structure. We know this as Bragg's Law of X-ray diffraction, and can be represented mathematically as such²: $n\lambda = 2d\sin\theta$, where n is an integer multiple of the wavelengths, and $2d\sin\theta$ is the path difference between two X-rays hitting and reflecting off of the crystal. Thus the formula represents constructive interference between multiple X-rays, leading to an increase in intensity.

While the Bragg experiment attempted to determine the lattice structure of crystals, we are attempting to use and confirm Bragg's Law in order to calculate specific X-ray wavelengths that are emitted from Molybdenum. The two X-ray types we are focusing on are known as K_α and K_β . K_α refers to the emission of a photon due to an electron shifting from the $n=2$ to $n=1$ energy level, and K_β comes from a shift from $n=3$ to $n=1$. Each arrives in the $n=1$ state or K-shell, hence the title "K." Using known lattice spacings of specific salt crystals, we can determine the wavelengths of these X-rays generated from Molybdenum.

Procedure (CK, AE, MN)

Our aim for this experiment is to acquire accurate wavelengths for K_α and K_β X-rays at various incident angles. Specifically, we are interested in angles that result in a sharp peak in

intensity and indicate constructive interference. In order to obtain our wavelength measurements for K_α and K_β , we use two different crystals: NaCl and LiF. In addition to this equipment, we use the Röntgengerät X-ray Apparatus, a powerful X-ray generating machine that allows us to adjust the incident angle of oncoming X-rays while also informing us of the waves' intensity with the use of a Geiger counter. The X-rays are created via a reverse photoelectric effect. When the electrons are fired from the tube chamber, they have kinetic energy. As they approach the Molybdenum nuclei, they lose their kinetic energy due to the electric field. The closer they are, the more energy they lose. This energy is converted into X-rays, and depending on how far the electrons are from the nucleus, there will be different intensities for different wavelengths. These radiations are called Bremsstrahlung radiations³, and it gives us the continuous curve in our data.

When placing each crystal in position, we take note of its orientation as we believe the specific face directed toward the x-rays may affect the Bragg reflection. In order to determine the optimal orientation of the block, we label two adjacent sides with a light pencil mark of an arrow pointing up to help us keep track of which position the crystal is in, which includes the rotation as well as up or down. We determine the optimal orientation based on the consistency and precision of the wavelengths across all orders of n for K_α and K_β x-rays.

When beginning data collection, we set the voltage (U) to 35.0 kV, the emission current (I) to 1.00 mA, the angular step width ($\Delta\theta$) to 0.1° , and we adjust the range of covered angles to have a lower limit of 2° and upper limit of 25° (using the COUPLED feature of the apparatus which adjusts the target and sensor accordingly). **The time the machine spends collecting data at each tenth of a degree (Δt) we adjust to 5 seconds.** Once the machine begins scanning, the computer instantaneously picks up and displays the measurements through the utilization of the X-ray Apparatus application. This

provides immediate feedback regarding the data collection and allows us to graphically interpret each of our trials regarding the incident/reflection angles and their corresponding X-ray intensities.

Results and Discussion (MN, AE, CK)

We realize that changing Δt affects only the count rate, or intensity, of the X-rays. This means that Δt does not affect the uncertainty in our data. For NaCl and LiF, we mostly use $\Delta t = 5$ to obtain our plots of count rate as a function of the incident angle, which can be seen in Figure 1 and Figure 2.

Bremsstrahlung Continuum Discussion

When the K_α and K_β peaks are removed, the figures have a main curve that has a peak at a certain wavelength, where the intensity is highest. This curve is called the Bremsstrahlung curve and is caused by the radiation emitted when the electrons lose kinetic energy as they pass through the Mo atom. This continuum is illustrated in Figure 5. The peak of this main curve occurs around 50 pm for both NaCl and LiF. See Figure 3 and Figure 4 for an illustration. This means that the type of crystal used does not affect the general shape of this main curve. Hence it is only dependent on the material used for X-ray emission which, in this case, is Molybdenum.

Our Data

The angles for the peaks of K_α and K_β are summarized in Table 1 for NaCl and Table 3 for LiF. Using Bragg's law of reflection, we calculate the wavelength of the characteristic peaks of K_α and K_β for different n values. The results are summarized in Table 2 for NaCl and Table 4 for LiF. Since LiF only has two sets of peaks between 2° and 25° , we use Bragg's law (using our

result for the wavelengths of K_α and K_β) to calculate the theoretical incident angles of the third-order ($n=3$) peaks for K_α and K_β). For K_α , the peak should be around 31.5° and K_β at approximately 27.7° . The peaks of K_α and K_β arise from constructive interference between the X-rays reflecting off the crystal. There are two features of these peaks. First, K_α always has a higher intensity than K_β . This is because K_α corresponds to a transition from $n=2$ to $n=1$ in the electron shells, while K_β corresponds to a transition from $n=3$ to $n=1$, which happens less often than the former. Second, the peaks for K_α come after the peaks for K_β . This can be explained by the higher photonic energy associated with the transition from $n=3$ to $n=1$, hence a shorter wavelength for K_β .

Uncertainties

The majority of our uncertainty comes from the measurement of the angle corresponding to each peak. To determine the uncertainty at a given angle associated with a peak, we take the full-width half max of each individual constructive interference peak, which are placed on the continuous curve. We choose the “bottom” of the peak to be where it meets the Bremsstrahlung curve. See Figure 6 for an illustration. The results for the uncertainty are in Table 2 for NaCl and Table 4 for LiF. We used these values to calculate the uncertainty in the wavelengths at each peak.

Since our equation, $n\lambda=2d\sin\theta$, is a single-variable function, we use section 4.1.3 from *Measurements and their Uncertainties* (Hughes and Hase) for the uncertainty in our wavelengths⁵. For $\sin(\theta)$, the error in the wavelength is $\alpha_\lambda = |\cos\theta|\alpha_\theta$, where α_θ is the error in the incident angle. For each calculated average wavelength, adding the relative uncertainties $(\frac{\alpha_\lambda}{\lambda})$ of each measurement, then converting back to absolute uncertainty (α_λ) gives us their uncertainties.

Discussion of Results

The expected value for K_{α} is 71.08 pm and K_{β} is 63.09 pm⁴.

For NaCl, our average wavelengths are 70 pm for K_{α} and 62.3 pm for K_{β} , which is off by 1.52% and 1.20% respectively compared to the expected values. Taking into account the relative uncertainty in our results, which are 1.43% and 1.44% respectively, the expected values are within the range of our results.

For LiF, the average wavelengths are 70.2 pm for K_{α} and 62.4 pm for K_{β} , which is off by 1.24 % and 1.09 % respectively compared to the expected values. Taking into account the relative uncertainty in our results, which are 0.85% and 0.96%, the expected values are not within the range of our results.

We find that the characteristic wavelengths, K_{α} and K_{β} , of Molybdenum are consistent when using NaCl, but are not when using LiF. We believe this is due to the limited number of data points for LiF since we use only data for $n=1$ and $n=2$, which decreases the overall uncertainty in its result.

Bibliography (CK)

- 1 Thomas, J. The birth of X-ray crystallography. Nature 491, 186–187 (2012).
- 2 Tipler, P.A. and Llewellyn, R.A. (2012) Modern Physics. 6th Edition, W. H. Freeman and Company, New York.
- 3 Nave, R. Bremsstrahlung X-rays, HyperPhysics,
<http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/xrayc.html>
- 4 LD Physics Leaflets, P6.3.3.1. Bragg reflection: diffraction of x-rays at a monocrystal.
- 5 Hughes, I., & Hase, T. P. A. (2010). Measurements and their uncertainties: A practical guide to modern error analysis. Oxford: New York.

Table 1. This data shows incident angles and corresponding intensities for both K_α and K_β with uncertainty in NaCl.

n	$\theta(K\alpha)^\circ$	$I(K\alpha) \text{ R/(1/s)}$	$\theta(K\beta)^\circ$	$I(K\beta) \text{ R/(1/s)}$
1	7.1 ± 0.3	1880.7	6.3 ± 0.2	949.2
2	14.4 ± 0.3	470.5	12.8 ± 0.4	214.7
3	22.1 ± 0.4	119.3	19.5 ± 0.3	63.2

Table 2. This data displays calculated wavelengths at each value of “n” in NaCl as well as the average for each X-ray.

n	$\lambda(K\alpha)$ pm	$\lambda(K\beta)$ pm
1	69.7 ± 0.3	61.9 ± 0.2
2	70.6 ± 0.3	62.5 ± 0.4
3	70.7 ± 0.4	62.6 ± 0.3
Average	70 ± 1	62.3 ± 0.9

Table 3. LiF result showing incident angle and corresponding intensities for both K_α and K_β with angle uncertainty.

n	$\theta(K\alpha)^\circ$	$I(K\alpha) \text{ R/(1/s)}$	$\theta(K\beta)^\circ$	$I(K\beta) \text{ R/(1/s)}$
1	10.0 ± 0.3	1679.2	8.9 ± 0.3	685.6
2	20.5 ± 0.3	322.6	18.1 ± 0.3	160.6

Table 4. LiF result showing calculated wavelengths for each value of “n” along with uncertainties for each and an average value.

n	$\lambda(K\alpha)$ pm	$\lambda(K\beta)$ pm
1	69.9 ± 0.3	62.3 ± 0.3
2	70.5 ± 0.3	62.5 ± 0.3
Average	70.2 ± 0.6	62.4 ± 0.6

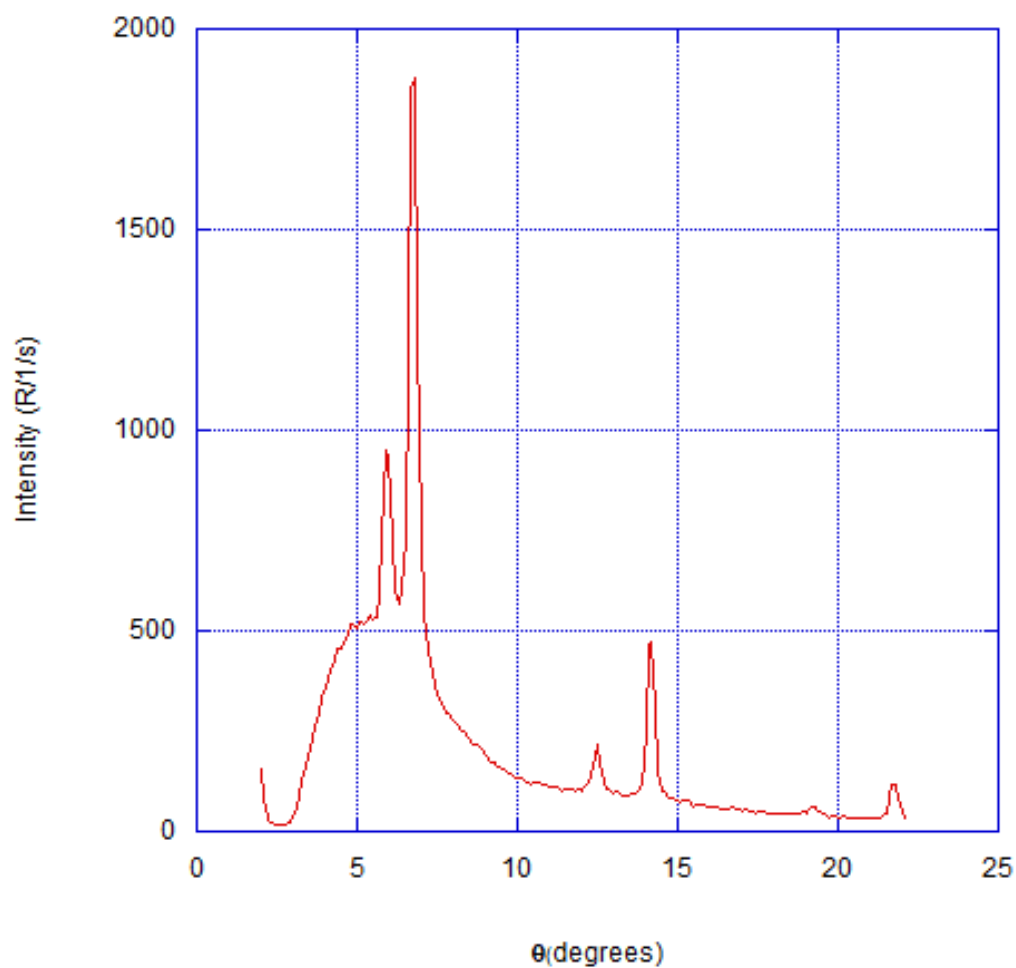


Figure 1. An Intensity vs Incident Angle plot for NaCl shows three distinct sets of peaks. Each set of peaks indicates the integer multiple “ n ” regarding constructive interference. The K_{β} always appears left of the corresponding K_{α} peak because of its inherently shorter wavelength.

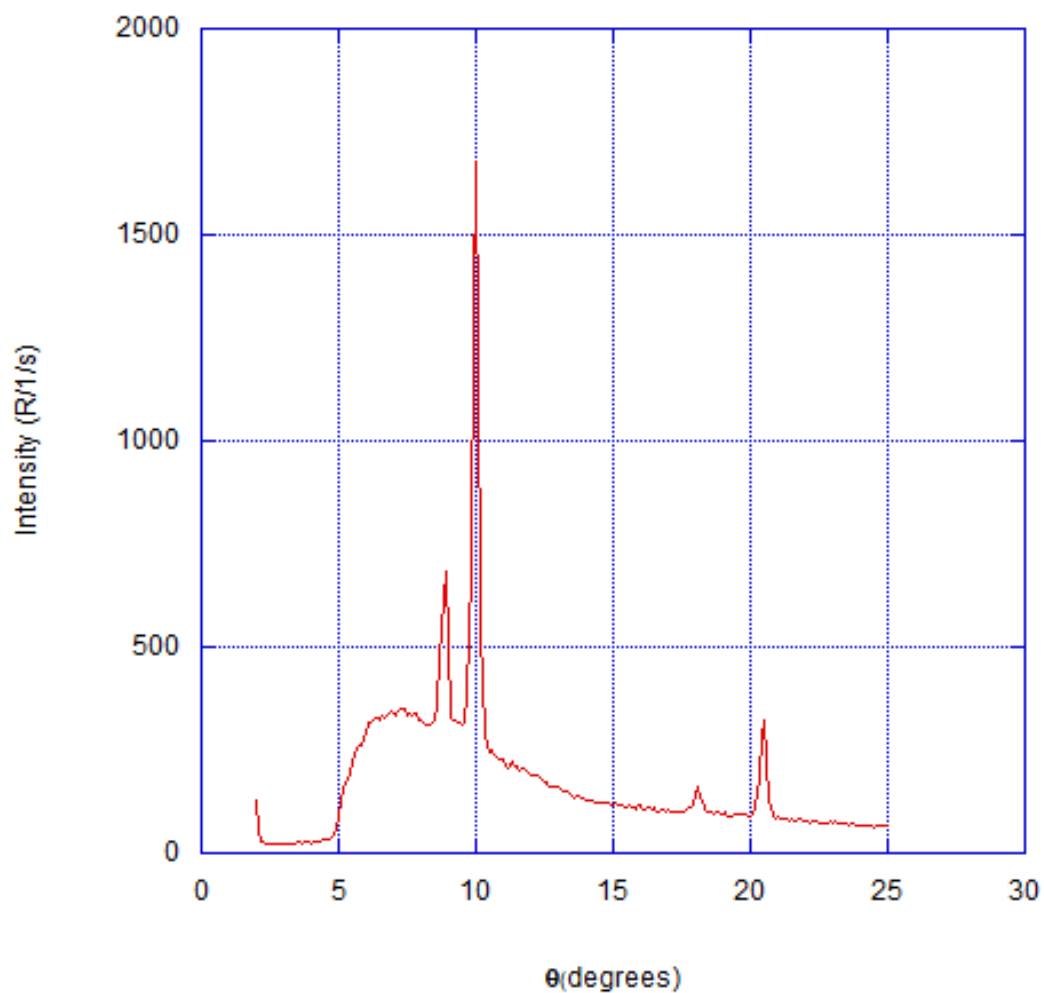


Figure 2. An Intensity vs Incident Angle plot for LiF shows two distinct sets of peaks. Each set of peaks, again, indicates the integer multiple “n” regarding constructive interference. Note that the $n=3$ set is not present, and would occur around 27.7° for K_β and 31.5° for K_α .

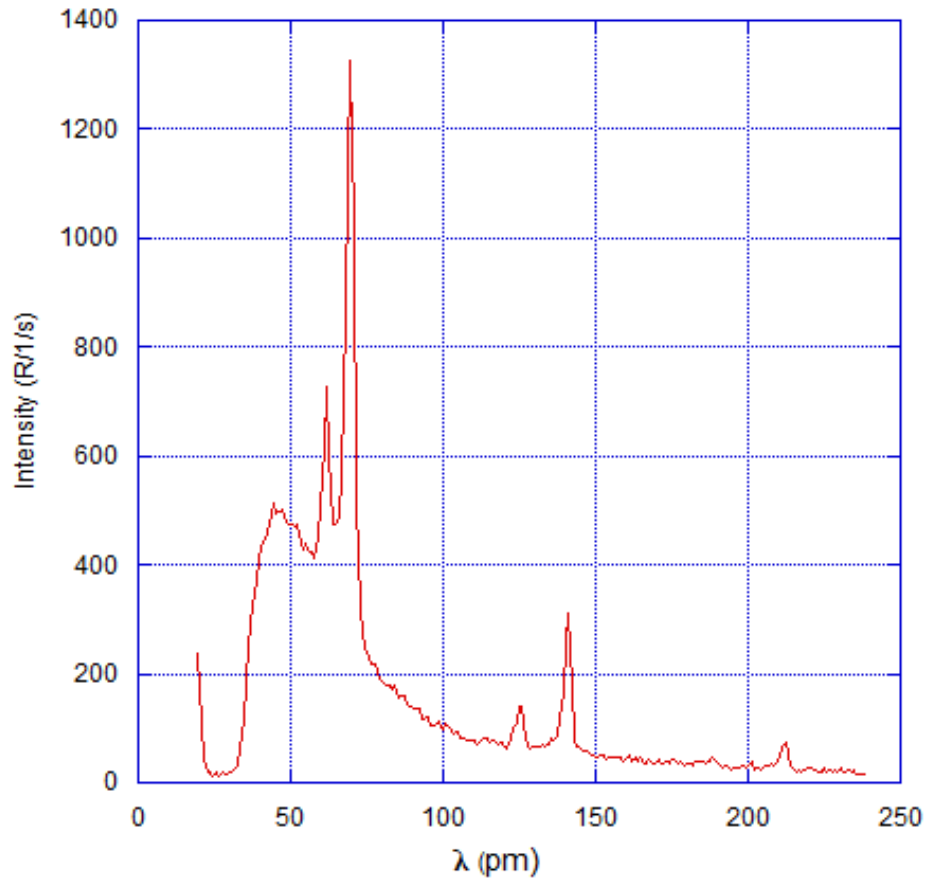


Figure 3. An Intensity vs Wavelength plot for NaCl. This attempts to illustrate that the shape of the plot, disregarding the peaks, is due to Bremsstrahlung radiation, which is illustrated in Figure 5. Notice the local maximum at 50 pm is consistent with LiF (see Figure 4) as well as with the Bremsstrahlung Continuum at 35 kV of potential.

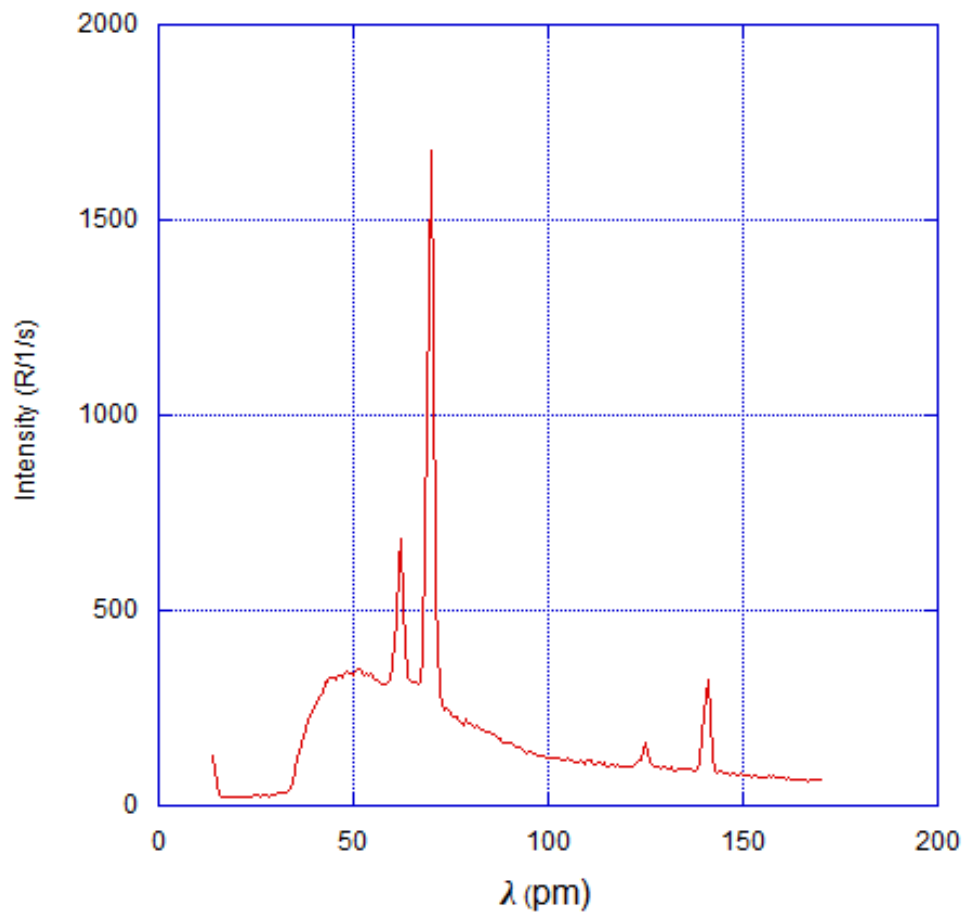


Figure 4. An Intensity vs Wavelength plot for LiF. This, again, attempts to illustrate that the shape of the plot, disregarding the peaks, is due to Bremsstrahlung radiation, which is illustrated in Figure 5. Notice the local maximum at 50 pm is consistent with NaCl (see Figure 3) as well as with the Bremsstrahlung Continuum using 35 kV of potential.

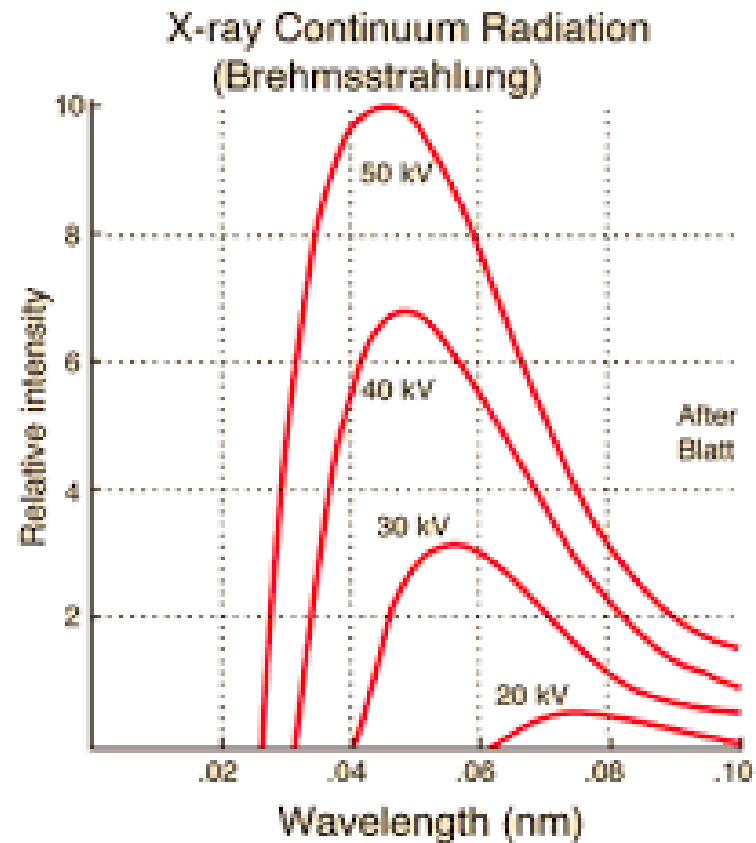


Figure 5. Author: R. Nave³. Illustrating the Bremsstrahlung Continuum, the shape of each curve is dependent upon the voltage used in generating the X-rays. This plot is used to explain the general shape of each plot we compute, ours being at 35 kV.

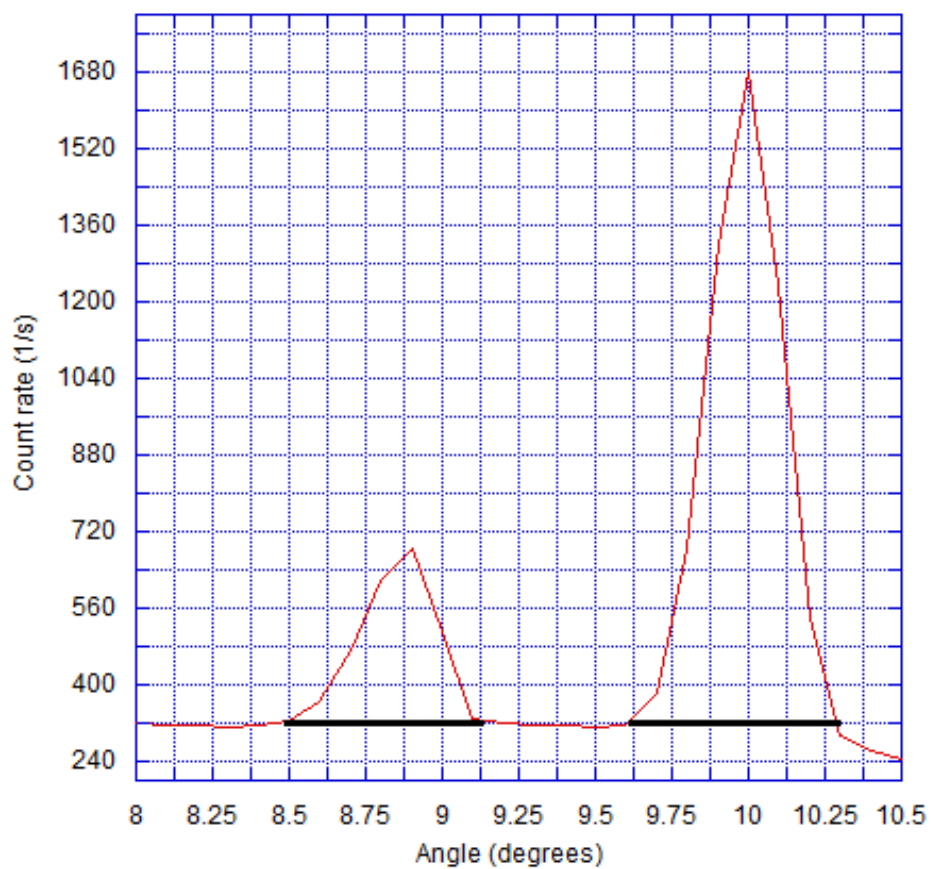


Figure 6. This plot illustrates the full-width half-max method of determining uncertainty at a given incident angle. The black horizontal lines indicate the “bottom” of each peak. We look at the width of the peak at the vertical midpoint of each peak to establish each uncertainty.